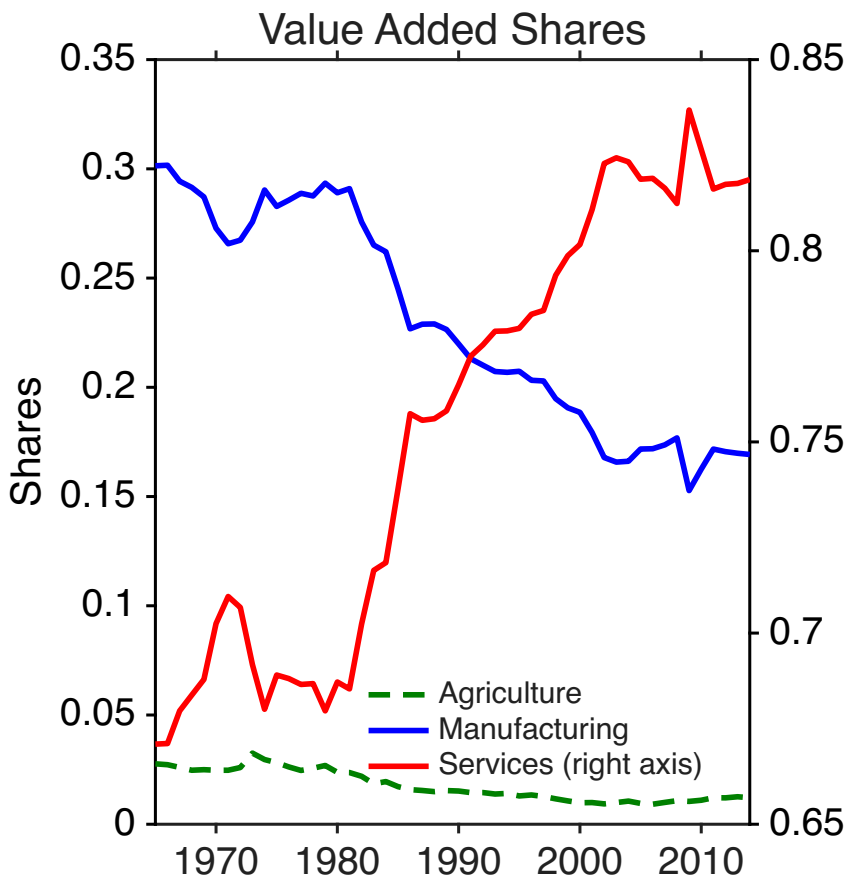
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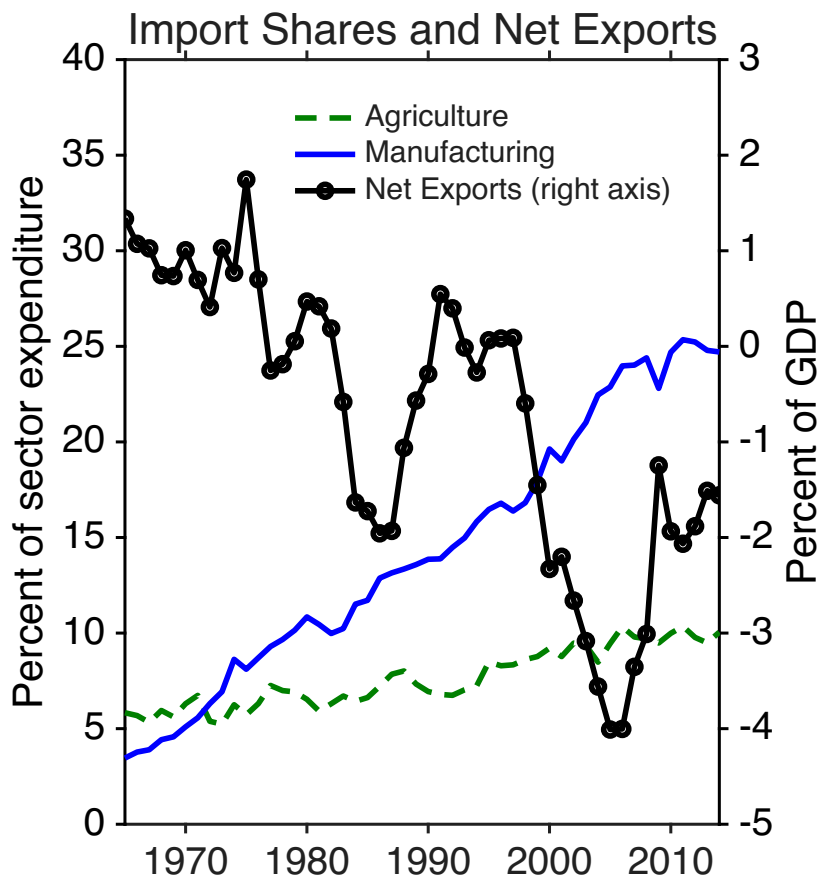
Mitchell Ballif

August 13, 2026

Introduction

Motivation





Introduction

Research question

How much have trade costs, working through changes in the trade deficit, affected the process of structural transformation in the United States?

(sectoral value-added shares)

Introduction

Literature

Structural change in a closed economy

- Technology story — Baumol (1967); Ngai & Pissarides (2007)
- Non-homothetic preferences — Kongsamut, Rebelo & Xie (2001); Boppart (2014); Comin, Lashkari & Mestieri (2021)

Structural change in an open economy

- Comparative advantage — Matsuyama (1992, 2009)
- Quantitative decompositions — Uy, Yi & Zhang (2013); Święcki (2017); Sposi (2019); Cravino & Sotelo (2019); Sposi, Yi & Zhang (2024)

Trade imbalances

- Imbalances $\rightarrow$ sectoral composition — Kehoe, Ruhl & Steinberg (2018)

Introduction

Contribution

1. Extended baseline SC model (trade imbalance + financial frictions)
2. Wrote math appendix (Solving and inverting equilibrium conditions)
3. Built LR Pannel using World IO Data
4. Took the model to Data (recovered shocks)

Model

Environment

- Time is discrete, $t = 1965, \ldots, 2014$.
- Two countries: US and the rest of the world, $i \in \mathcal{I} \equiv \{US, ROW\}$.
- Three sectors in each country, all tradable, $j \in \mathcal{J} \equiv \{a, m, s\}$.
- Representative household in each country:
  - Country i endowed with $L_{i,t}$ units of homogeneous labor in every t and $K_{i,0}$ units of homogeneous physical capital.
  - Access to international financial markets by means of buying and selling one-period bonds with portfolio adjustment costs.
  - Born with zero NFA, $B_{i,0} = 0$.

Model

The Household's Dynamic Problem

The representative household in country i chooses consumption, investment, capital, and bonds to maximize:

$$\max \sum_{t=0}^{\infty} \delta^t L_{i,t} \ln\left(\bar{C}_{i,t}\right)$$

subject to

$$E_{i,t}^{C} + E_{i,t}^{X} + B_{i,t+1} + \frac{\phi_{i,t}}{2}\left(B_{i,t+1} - \bar{B}_i\right)^2 = w_{i,t} L_{i,t} + \chi_{i,t} r_{i,t} K_{i,t} + R_t B_{i,t} + T_{i,t}$$

$$K_{i,t+1} = (1-d) K_{i,t} + \left(X_{i,t}\right)^{\sigma} \left(K_{i,t}\right)^{1-\sigma}$$

where $\bar{C}_{i,t}$ is defined implicitly (CLM, 2017):

$$\sum_{j \in \mathcal{J}} \left(\mu_{i,t}^{j}\right)^{\frac{1}{\psi}} \left(\frac{C_{i,t}}{L_{i,t}}\right)^{\frac{1-\psi}{\psi}\varepsilon^{j}} \left(\frac{C_{i,t}^{j}}{L_{i,t}}\right)^{\frac{\psi-1}{\psi}} = 1$$

and $\psi > 0$, $\mu_{i,t}^{j} > 0$, and $\sum_j \varepsilon^j = 1$.

Model

Methodology

1. Invert the model and find shocks.

2. Given parameters, $K_{i,1}$, $B_{i,1} = 0$, and paths for $\{\mu^j_{i,t}, T^j_{i,t}, \tau^j_{ih,t}, \chi_{i,t}, \phi_{i,t}\}$, find paths of prices, quantities, trade shares and bonds such that all equilibrium conditions hold.

3. Compare results against counterfactual equilibria with fixed trade costs and financial frictions.

Model

Equilibrium Conditions

Tag	Equation / Recovery Formula	Description
(C1)	$P^C_{i,t} \equiv \left[\sum_j \mu^j_{i,t} \left(\frac{C_{i,t}}{L_{i,t}}\right)^{(1-\psi)(\varepsilon^j-1)} (P^j_{i,t})^{1-\psi}\right]^{\frac{1}{1-\psi}}$	Consumption price index
(X1)	$P^X_{i,t} \equiv \left[\sum_j \upsilon^j_i (P^j_{i,t})^{1-\zeta}\right]^{\frac{1}{1-\zeta}}$	Investment price index
(F1)	$c^j_{i,t} = \kappa^j_i \left[(r^{\varphi_i}_{i,t} w^{1-\varphi_i}_{i,t})^{\beta^j_i} \prod_m (P^m_{i,t})^{\nu^{j,m}_i}\right]$	Sectoral unit cost
(T1)	$\pi^j_{ih,t} = \frac{T^j_{h,t}(c^j_{h,t}\tau^j_{ih,t})^{-\theta}}{\Phi^j_{i,t}}$	Trade share (Gravity)
(E1)	$\frac{E^C_{i,t+1}}{E^C_{i,t}} = \frac{L_{i,t+1}}{L_{i,t}} \frac{\widetilde{\phi}_{i,t+1}}{\widetilde{\phi}_{i,t}} \frac{\delta R_{t+1}}{1+\phi_{i,t}(B_{i,t+1}-\bar{B}_i)}$	Bond Euler equation
(K1)	$\frac{R_{t+1}}{1+\phi_{i,t}(B_{i,t+1}-\bar{B}_i)} P^X_{i,t} = \chi_{i,t+1} r_{i,t+1} + (1-d) P^X_{i,t+1}$	Capital Euler ($\sigma = 1$)
(INV1)	$T^j_{i,t} = \pi^j_{ii,t} \left(\Gamma \frac{c^j_{i,t}}{P^j_{i,t}}\right)^{\theta}$	Productivity inversion
(INV2)	$\tau^j_{ih,t} = \frac{P^j_{i,t}}{P^j_{h,t}} \left(\frac{\pi^j_{hh,t}}{\pi^j_{ih,t}}\right)^{\frac{1}{\theta}}$	Trade cost inversion
(INV3)	$\phi_{R,t} = \frac{1}{B_{R,t+1}-\bar{B}_R} \left[\frac{E^C_{R,t}}{E^C_{R,t+1}} \frac{L_{R,t+1}}{L_{R,t}} \frac{\widetilde{\phi}_{R,t+1}}{\widetilde{\phi}_{R,t}} \delta R_{t+1} - 1\right]$	Financial friction inversion
(INV4)	$\chi_{i,t+1} = \frac{1}{r_{i,t+1}} \left[\frac{R_{t+1}}{1+\phi_{i,t}\left(B_{i,t+1}-\bar{B}_i\right)} P^X_{i,t} - (1-d) P^X_{i,t+1}\right]$	Capital wedge inversion

Recovering parameters

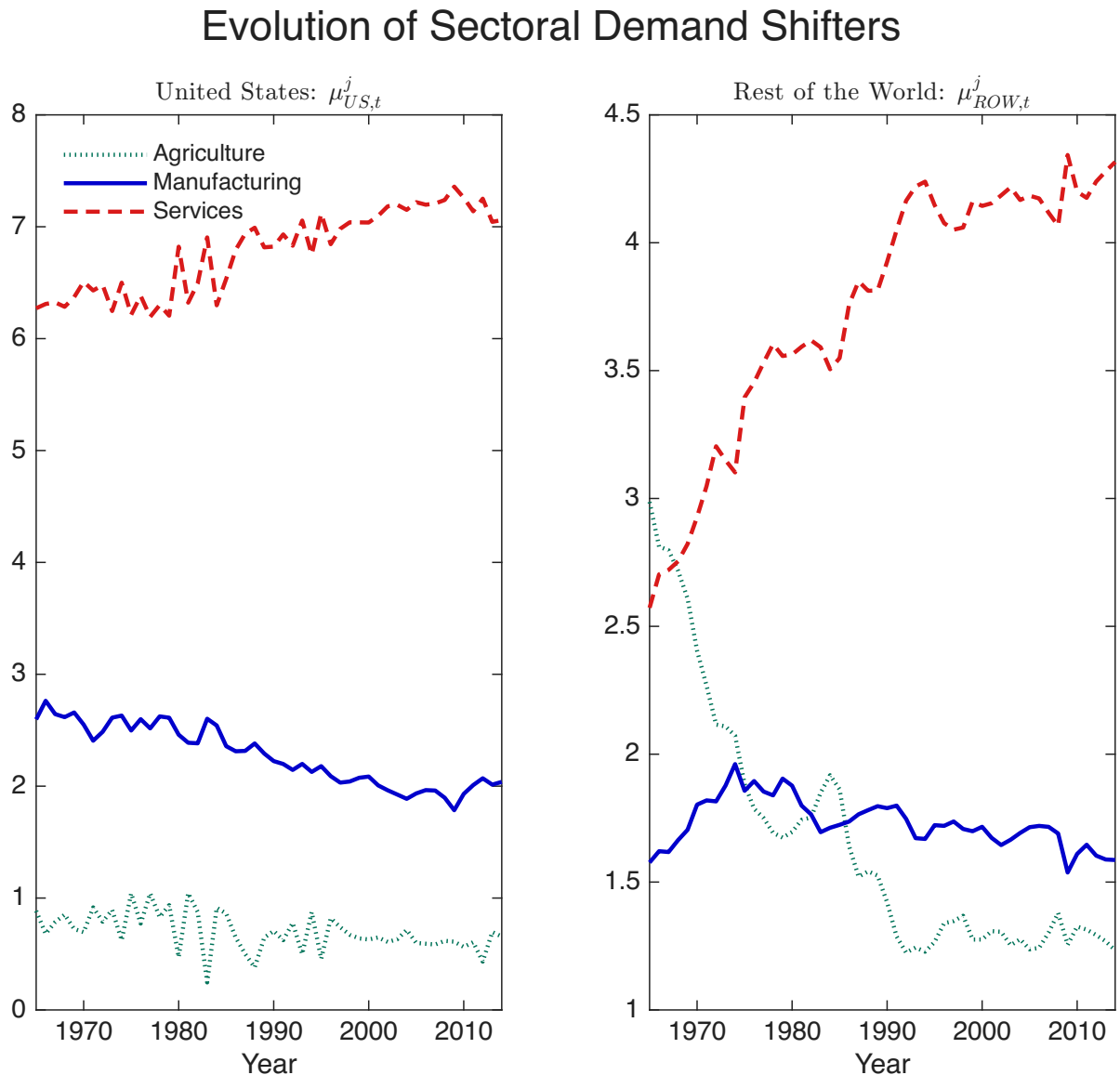
Data

Source	Years	What it gives
Long-Run WIOD, v1.1	1965–2000	Input–output tables, gross output, value added, output deflators
WIOD 2016 release	2000–2014	The same, at finer country and sector detail
GGDC Productivity Level DB	2005	Benchmark sectoral PPPs, so prices compare across countries ▸
Penn World Table 11.0	1965–2014	Employment, capital, investment prices, exchange rates

Spliced at 2000 into one panel: two countries (U.S. and rest of world), ▸ three sectors (agriculture, manufacturing, services), 1965–2014. Levels come from the long-run release through 2000 and growth rates from the 2016 release after.

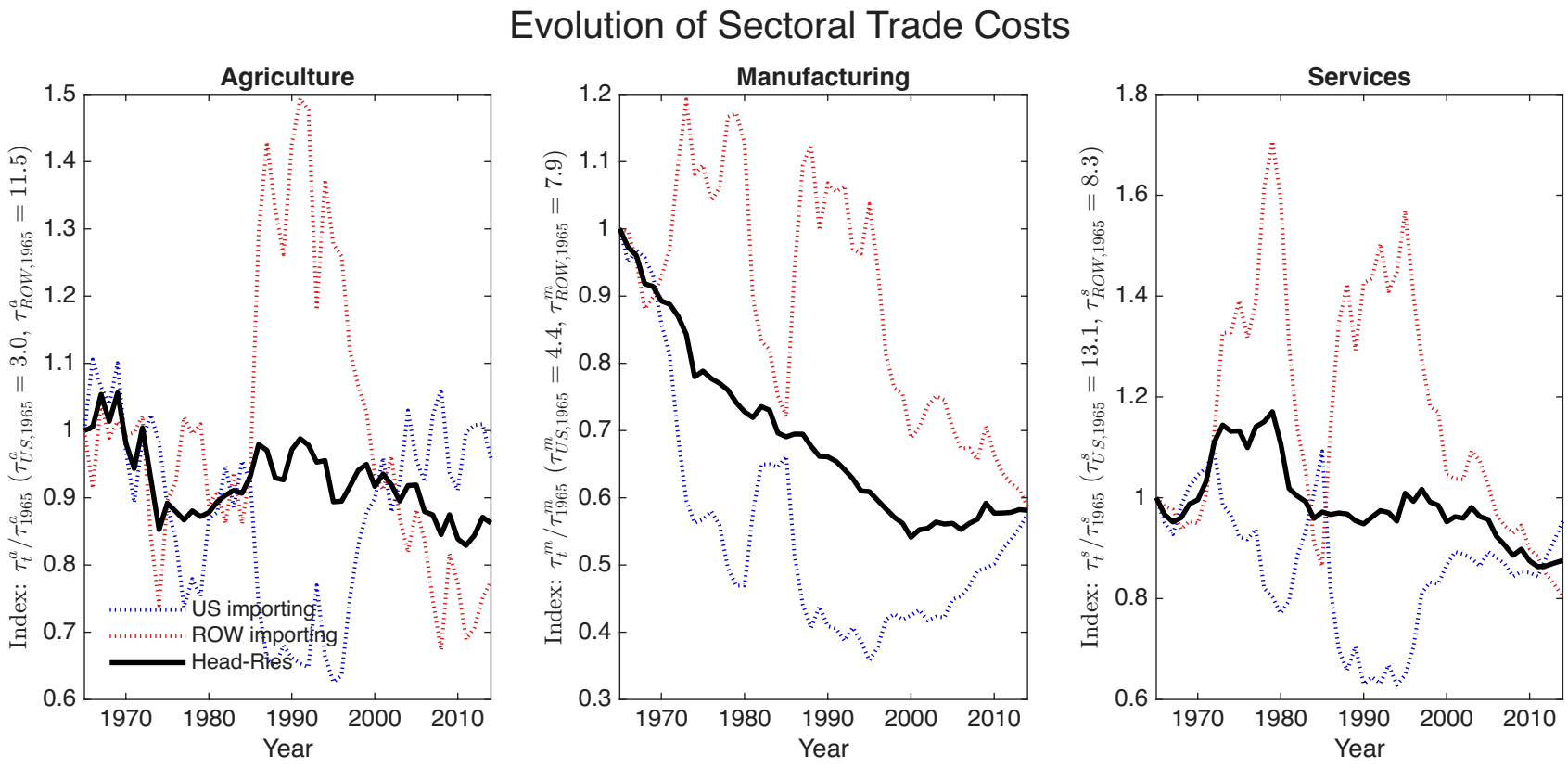
Recovering parameters

Demand Shifters



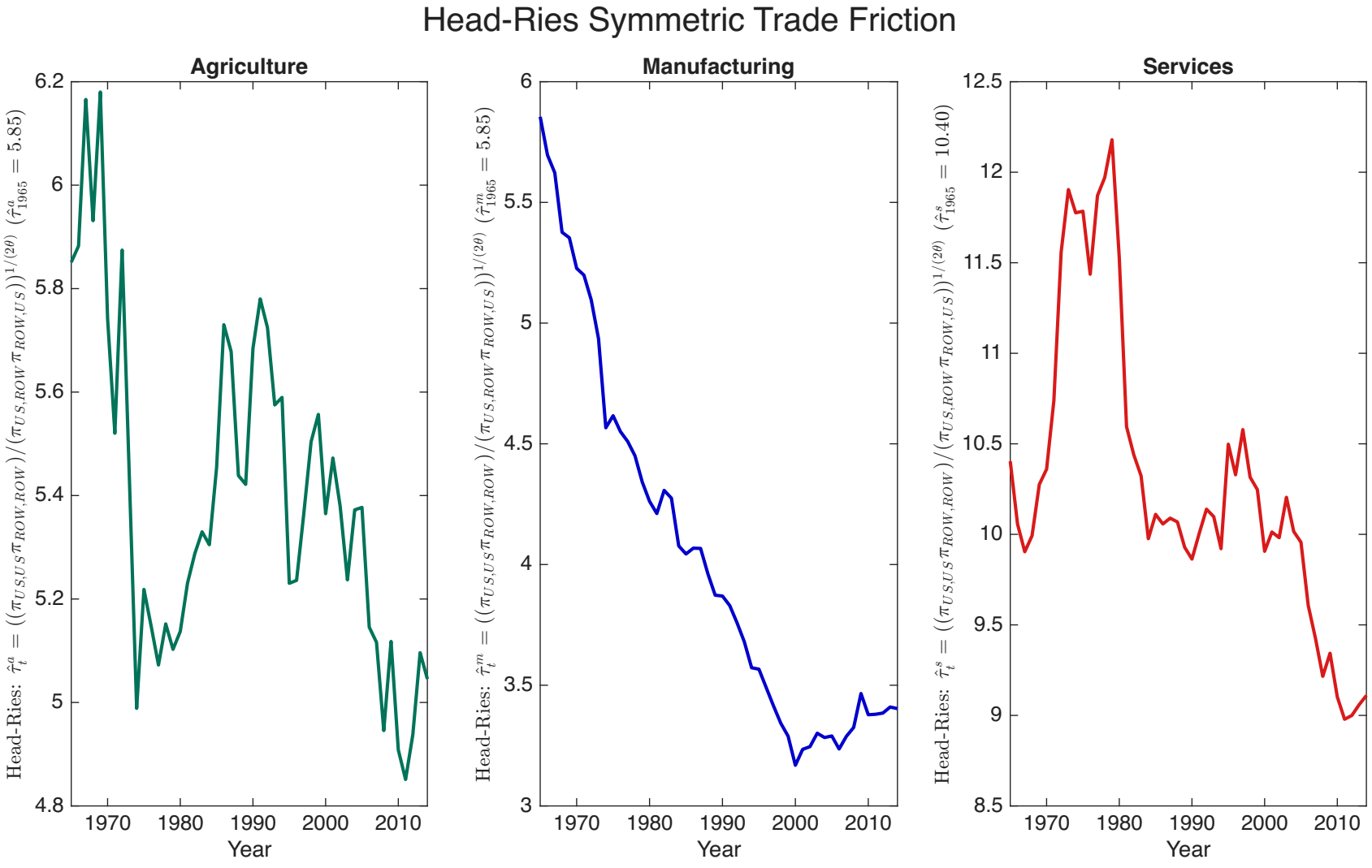
Recovering parameters

Trade Costs



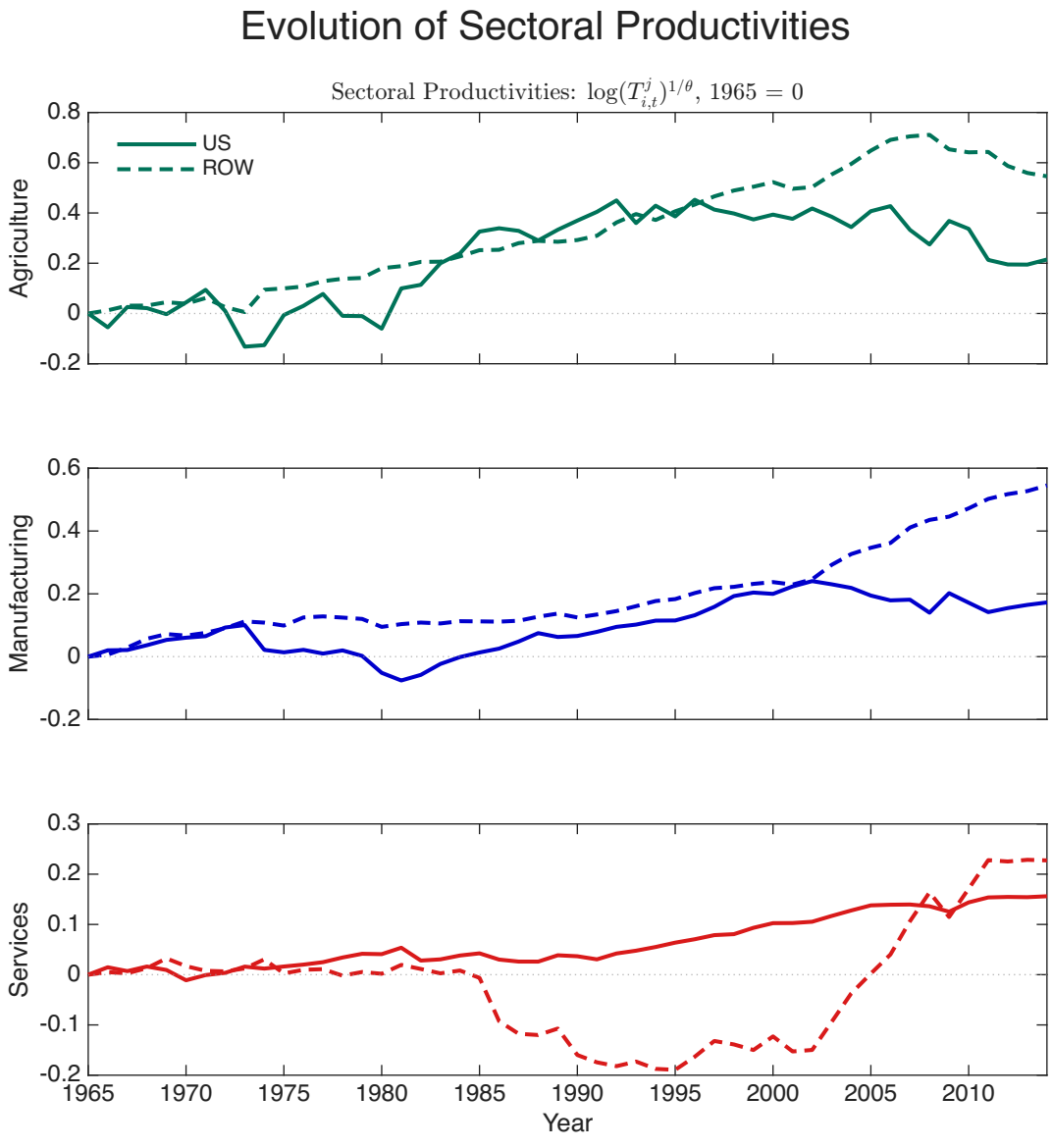
Recovering parameters

Head-Ries Index



Recovering parameters

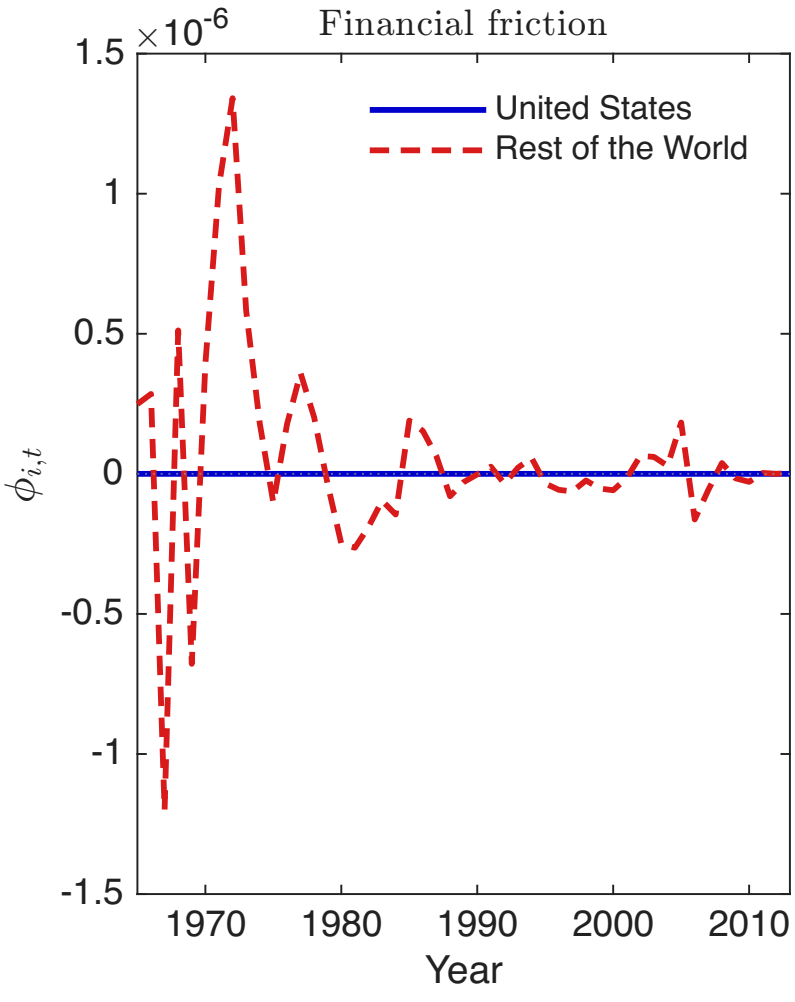
Productivities

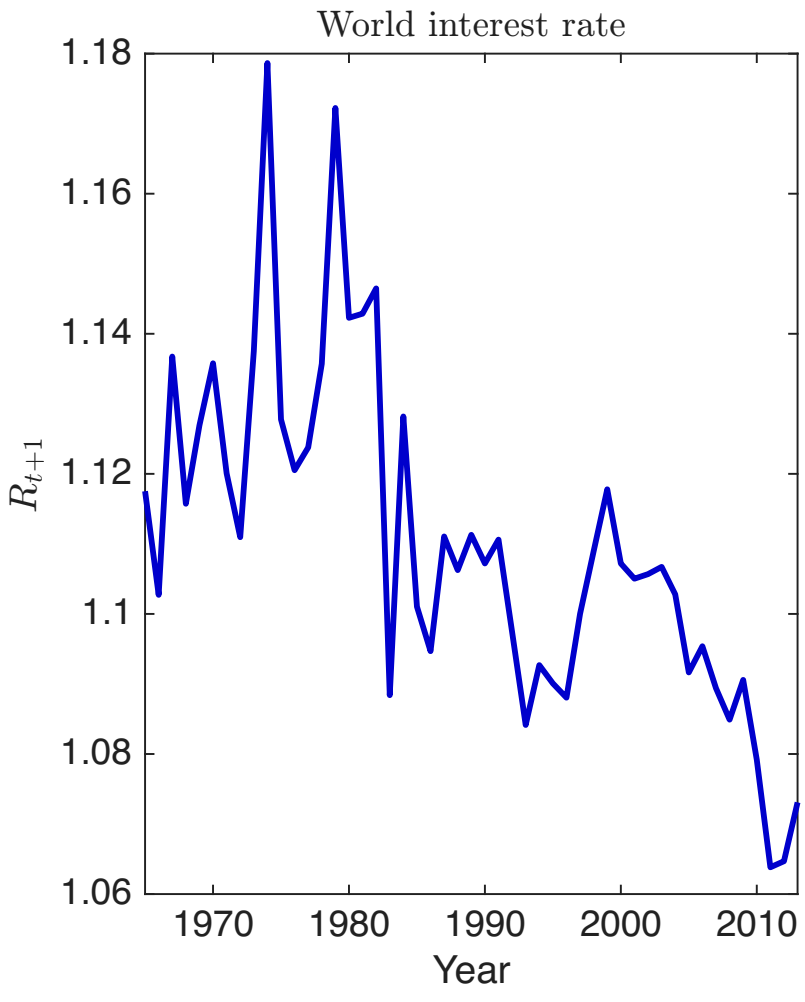


Recovering parameters

Financial Friction

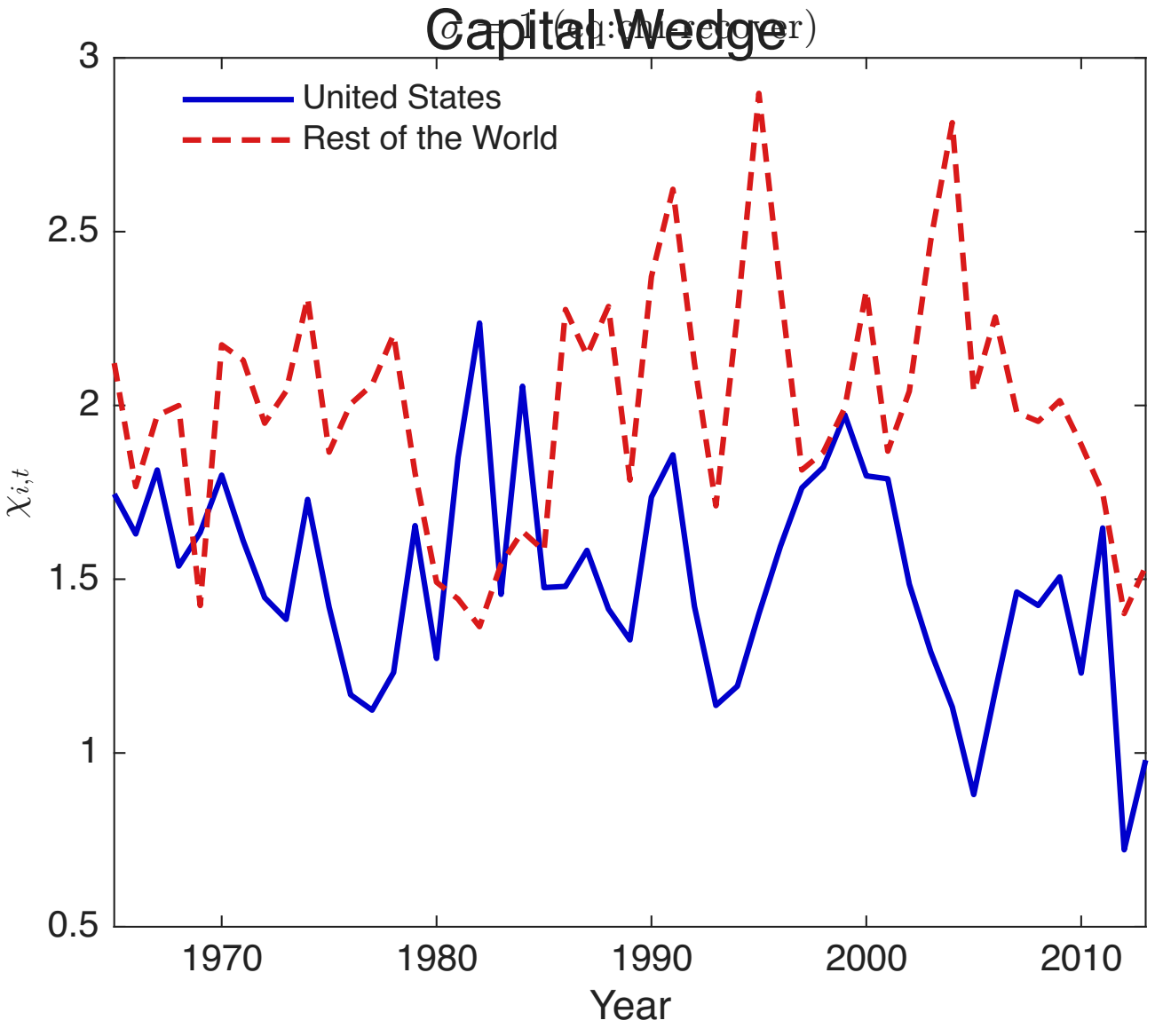
Financial Friction (world rate: mean 1.110, range 1.064 to 1.179)





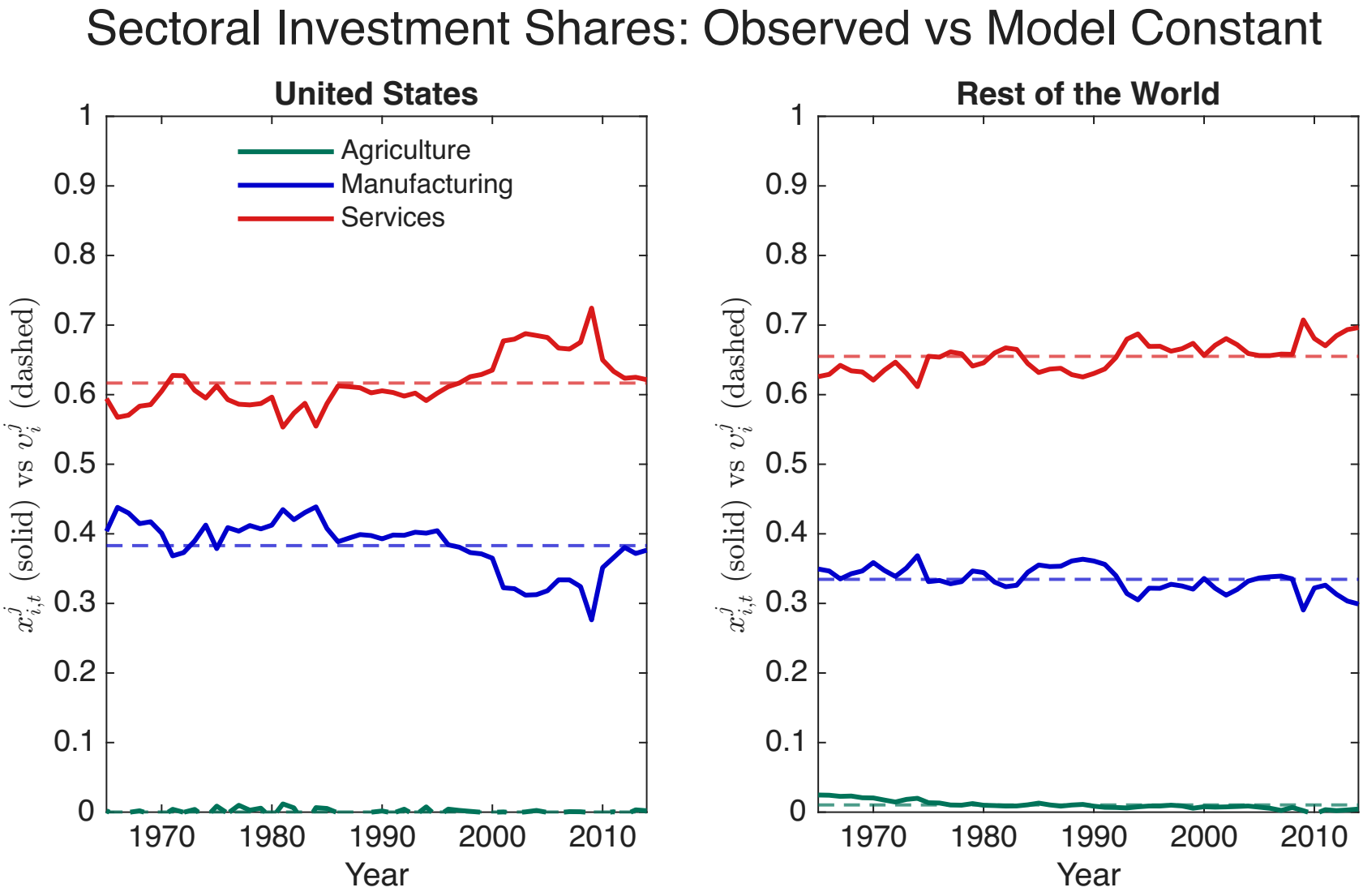
Recovering parameters

Investment Efficiency



Recovering parameters

Investment Shares



Counterfactuals

TO ADD.

Conclusion

TO ADD.

Thank you

Appendix

Sectoral Prices and Trade Shares (Eaton-Kortum)

- **Ideal Price Index:** Sectoral prices in each j are given by

$$P_{i,t}^{j} = \Gamma \left(\Phi_{i,t}^{j} \right)^{-\frac{1}{\theta}}, \quad \text{where} \quad \Phi_{i,t}^{j} = \sum_{h \in \mathcal{I}} T_{h,t}^{j} \left(c_{h,t}^{j} \tau_{ih,t}^{j} \right)^{-\theta}$$

- **Unit Cost:** The minimum cost of the input bundle $c_{i,t}^{j}$ is

$$c_{i,t}^{j} = \kappa_{i}^{j} \left[\left(r_{i,t} \right)^{\varphi_i} \left(w_{i,t} \right)^{1-\varphi_i} \right]^{\beta_i^j} \left[\prod_{k \in \mathcal{J}} \left(P_{i,t}^{k} \right)^{\nu_i^{j,k}} \right]^{1-\beta_i^j}$$

- **Trade Shares:** Define $E_{i,t}^{j} \equiv P_{i,t}^{j} Q_{i,t}^{j}$ as total expenditure on sector j. The share of expenditure on goods imported from h is:

$$\pi_{ih,t}^{j} \equiv \frac{E_{ih,t}^{j}}{E_{i,t}^{j}} = \frac{T_{h,t}^{j} \left(c_{h,t}^{j} \tau_{ih,t}^{j} \right)^{-\theta}}{\Phi_{i,t}^{j}}$$

This is the multisector version of the gravity equation.

◀ back

Appendix

Sectoral price levels: construction

The goal is a price for each country, sector and year that is comparable *across* countries, not only over time.

1. **Deflators.** Gross output price indices from both WIOD releases. The long-run release is in USD, so it is multiplied by the *2005* exchange rate to put it in local currency; the 2016 release is already in local currency.
2. **Splice.** Chain-link at 2000 — long-run levels through 2000, 2016 growth rates after — then normalize to 2005 within each country and sector, so $\tilde{\pi}_{i,c,2005} = 1$.
3. **Benchmark.** Take the 2005 gross-output PPP for each country and sector from PLD, $\text{PPP}_{i,c}$, in local currency per USD. It is a gross-output-weighted geometric mean of the product-level PPPs.
4. **Combine,** holding the exchange rate at its 2005 value:

$$P_{i,c,t} = \tilde{\pi}_{i,c,t} \cdot \frac{\text{PPP}_{i,c}}{e_{c,2005}}$$

At 2005 the level is the sector's PPP relative to the United States; in every other year it moves only with that country's own deflator. Fixing the exchange rate keeps the nominal dollar cycle out of the cross-country comparison. For the United States PPP $= e = 1$, so the series is a pure within-country price index.

◀ back

Appendix

Sectoral price levels: aggregating to ROW

The model has two countries, so member price levels have to be combined into one. Gross-output-weighted geometric mean of the *levels*, computed within each year:

$$\ln P_{i,\mathrm{ROW},t} = \sum_{c \in \mathrm{ROW}} w_{c,t} \ln P_{i,c,t}, \qquad w_{c,t} = \frac{Y_{c,t}}{\sum_{c' \in \mathrm{ROW}} Y_{c',t}}$$

- A mean of levels, not a chained index of changes: a cheap member lowers the ROW price level as soon as it gains share.
- A chained index reads only price *changes*, so an entrant contributes nothing on arrival and its low level is never picked up. Using one drove measured ROW services productivity negative.
- Everything is aggregated to two countries first, and the disturbances are recovered from the aggregated data, so no member-level parameter is pooled. Bilateral trade costs follow directly from the aggregated prices and trade shares, $\tau^{j}_{ih,t} = \dfrac{P^{j}_{i,t}}{P^{j}_{h,t}} \left(\dfrac{\pi^{j}_{hh,t}}{\pi^{j}_{ih,t}} \right)^{1/\theta}$.

◀ back

Appendix

Market Clearing Conditions

- **Nontradable Goods & Factor Markets:**

$$C_{i,t}^{j} + X_{i,t}^{j} + \sum_{k \in \mathcal{J}} D_{i,t}^{k,j} = Q_{i,t}^{j} \quad \text{for all } j \in \mathcal{J}$$

$$\sum_{j \in \mathcal{J}} L_{i,t}^{j} = L_{i,t} \quad \text{and} \quad \sum_{j \in \mathcal{J}} K_{i,t}^{j} = K_{i,t}$$

- **Value of Production:** Let $Y_{i,t}^{j}$ denote the gross value of production.

$$Y_{i,t}^{j} = \sum_{h \in \mathcal{I}} \pi_{hi,t}^{j} E_{h,t}^{j} \quad \text{for all } j \in \mathcal{J}$$

- **Country-Specific Resource Constraint:**

$$B_{i,t+1} - R_t B_{i,t} = \sum_{j \in \mathcal{J}} Y_{i,t}^{j} - E_{i,t}^{j}$$

- **International Financial Markets Clear:**

$$\sum_{i \in \mathcal{I}} B_{i,t+1} = 0 \quad \text{for all } t$$

◀ back

Appendix

Calibration

θ	trade elasticity	4
η	elasticity across varieties	2
ψ	elasticity across sectors	0.57
ζ	investment elasticity	1
σ	capital adjustment	0
δ	discount factor	0.95
d	depreciation	0.05

ε^j from Comin, Lashkari and Mestieri (2021); φ_i from Caselli and Feyrer (2007); β_i^j and $\nu_i^{j,m}$ are sample averages.

◀ back